

Ju O Kim

AI Research Engineer · Physics-Informed ML for Industrial Systems

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Profile

AI Research Engineer and Computer Engineering Ph.D. student working at the intersection of physics-informed machine learning and industrial systems. Research spans neural operators, acoustic tomography, and physics-informed Bayesian optimization; engineering spans multimodal sensing, anomaly detection, and production scheduling. Author or co-author of 20 publications, including five first-author SCI/SCIE journal papers.

Core Expertise

Scientific & Industrial ML: **PyTorch, neural operators, Gaussian Processes** - BoTorch, GPyTorch, XGBoost, Optuna, Ray

Signal & Sensing: **Acoustic, thermal, UWB radar** - beamforming (MVDR/DAS/MUSIC), GCC-PHAT/TDOA, OpenCV

Systems: **Python, FastAPI, Docker** - Celery, Redis, PostgreSQL, TimescaleDB, ONNX

Agents & XAI: **LangGraph, MCP, Ollama** - deterministic guardrails, human-in-the-loop, SHAP, DoWhy, causal-learn

Languages: **Python** - TypeScript, SQL, MATLAB | Korean (native), English (professional working proficiency)

Experience

AI Research Engineer, R&D Division - JS SYSTEM (Manufacturing AI Solutions)

Feb 2025 - Present

- Develop end-to-end AI systems spanning model development, data pipelines, APIs, deployment, and technical coordination.
- Developed an AutoML workflow combining model screening, hyperparameter optimization, and ensemble construction.
- Built acoustic anomaly-detection and on-prem multi-agent systems with local LLMs, deterministic guardrails, and human-in-the-loop control.
- Delivered REST APIs and real-time monitoring dashboards across the ML lifecycle.

Researcher, R&D Department - Seongwoo Instruments

Feb 2023 - Mar 2024

- Built a real-time micro-seismic monitoring and isolation system for superconducting seismic-wave detection under a government-funded national R&D program, in collaboration with the University of Maryland Physics Lab, including a 6-month on-site assignment at UMD.
- Designed digital filters and SVM/Random Forest classifiers for seismic-vs-noise discrimination; developed the MATLAB ISOLATOR module for real-time vertical-motion control.
- Engineered a high-throughput real-time data-streaming pipeline on a hybrid MATLAB-Python framework.

Commissioned Researcher - DGIST (Daegu Gyeongbuk Institute of Science & Technology)

Mar 2020 - Dec 2021

- Developed vanishing-point-based calibration for a three-viewpoint thermal camera with wide-baseline ($>180^\circ$ / $\sim 240^\circ$) view separation; reduced reprojection error to 0.58-0.66 px without correspondence matching (published in *Journal of Signal Processing Systems*, Springer).

Graduate Researcher, NRF Research Project - Keimyung University

2020 - 2022

- Designed a DNN architecture for depth-camera distortion correction (UAV/AMR perception), trained on real-world datasets spanning varied weather and lighting.

Selected Engineering Systems

APPLIED ENGINEERING AI Production Scheduler. Developed and validated a production-scale scheduling system for manufacturing workflows, outperforming a legacy planning baseline while satisfying all hard constraints.

PERSONAL R&D - OPEN/SYNTHETIC DATA GPO - Generative Process Optimizer. On an open 564-cycle injection-molding benchmark (n=20), Bayesian optimization reached 90% convergence in 14.8 evaluations versus 35.1 for random search; a physics-informed prior reduced that to 8.8.

R&D PROTOTYPE - OPEN/SYNTHETIC DATA Multi-Agent Manufacturing QC. Eleven agents cross-examine ML-tool evidence, while deterministic guardrails and human review own safety decisions. Runs on-prem with Ollama and is covered by 1,150 passing tests.

PERSONAL RESEARCH - PUBLIC DCASE/MIMII Acoustic Vision System. Adapts an 87M-parameter audio transformer with 0.68%-1.35% trainable parameters depending on machine type; self-evaluated under the DCASE 2020 Task 2 protocol at mean AUC ~0.94 across Fan, Pump, Slider, and Valve.

Selected Research & Publications

Neural acoustic tomography, *Measurement Science and Technology* (2026). Single-pass 3D reconstruction ran ~950x faster per scene and improved IoU by 51.0% over a generic encoder; a topology study found that the standard Eikonal prior reduced IoU by 9.3% overall.

FFT-free neural operator, *Applied Sciences* (2026). Achieved 4.6-16.4x faster inference than FNO/MscaleFNO and a 19.6% out-of-distribution gap, 2.6x lower than FNO.

Non-contact sensing and thermal calibration. Published a UWB respiratory classifier with 94.3% accuracy and 0.969 AUC (*Electronics*, 2025), and wide-baseline thermal-camera calibration with 0.58-0.66 px reprojection error (*Journal of Signal Processing Systems*, 2024).

Publication record. Five first-author SCI/SCIE journal papers; 20 publications total. Full list and paper links: slothisjp.github.io/#publications.

Education

Ph.D., Computer Engineering (Artificial Intelligence) - Keimyung University, Daegu 2026 - Present
Advisor: Prof. Deokwoo Lee.

M.Eng., Computer Engineering (Artificial Intelligence) - Keimyung University, Daegu 2020 - 2025
GPA 4.12 / 4.5. Thesis: *Deep Learning Approaches for Non-Contact Biomedical Signal Processing*. Advisor: Prof. Deokwoo Lee.

B.Eng., Computer Engineering - Keimyung University, Daegu 2014 - 2020
Includes mandatory military service (2014-2016).

Awards

Best Paper Award, Korean Signal Processing Society (2023).